

Topic P4: Magnetism and magnetic fields

P4.1 Magnets and magnetic fields

Summary

Having an understanding of how charge can be generated and its effects, we can now consider the link between movement of charge and magnetism. To begin, learners will investigate magnets and magnetic fields around magnets and current-carrying wires.

Underlying knowledge and understanding

Learners should have been introduced to magnets and the idea of attractive and repulsive forces. They should have an idea of the shape of the fields around bar magnets. Learners are expected to have an awareness of the magnetic effect of a current and electromagnets.

Common misconceptions

Common misconceptions that learners have include the idea that larger magnets will always be stronger magnets. They also have difficulty understanding the concept of field line density being an indicator of field strength. Learners often do not know that the geographic and magnetic poles are not located in the same place.

Tiering

Statements shown in **bold** type will only be tested in the Higher Tier papers. All other statements will be assessed in both Foundation and Higher Tier papers.

Topic content		Opportunities to cover:		Practical suggestions
Learning outcomes	To include	Maths	Working scientifically	
P4.1a describe the attraction and repulsion between unlike and like poles for permanent magnets	diagrams of magnetic field patterns around bar magnets to show attraction and repulsion		WS1.1b, WS1.2a, WS1.2b, WS2a, WS2b	Use of suspended magnets to show attraction and repulsion.
P4.1b describe the difference between permanent and induced magnets				
P4.1c describe the characteristics of the magnetic field of a magnet, showing how strength and direction change from one point to another	diagrams to show how the strength of the field varies around them and ways of investigating this	M5b	WS1.1b, WS1.2a, WS1.2b, WS2a, WS2b, WS2c	Plotting of magnetic fields around different shaped magnets.
P4.1d explain how the behaviour of a magnetic (dipping) compass is related to evidence that the core of the Earth must be magnetic				

P4.1e	describe how to show that a current can create a magnetic effect and describe the directions of the magnetic field around a conducting wire			WS1.1b, WS1.2a, WS1.2b, WS2a, WS2b, WS2c	Investigation of the magnetic field around a current-carrying wire using plotting compasses.
P4.1f	recall that the strength of the field depends on the current and the distance from the conductor		M1c		
P4.1g	explain how solenoid arrangements can enhance the magnetic effect		M1c	WS1.1b, WS1.2a, WS1.2b, WS2a, WS2b, WS2c, WS2d	Investigation of the magnetic field around a current-carrying solenoid using plotting compasses. Investigation of the factors that can affect the magnetic effect e.g. number of turns, and length.